

High-Availability Hospital Campus Network

Network Design and Implementation Documentation

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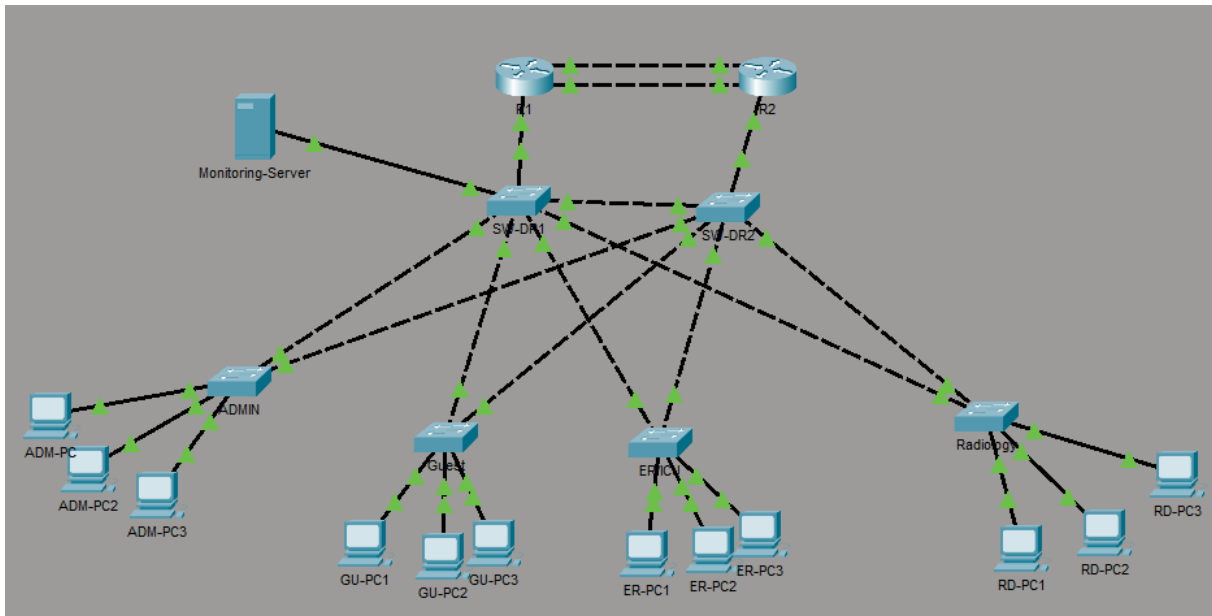
1. Project Overview

This project is a simulated hospital campus network built in Cisco Packet Tracer to demonstrate enterprise-level network design centered on high availability, dynamic routing, and departmental security segmentation. The design assumes that network downtime and uncontrolled access between departments are unacceptable in a healthcare environment, and every major design decision follows from that constraint.

The network was not only built but actively tested under simulated failure conditions — a core router shutdown, a primary link failure, and a switch-loop scenario — to verify that redundancy and security policy hold up in practice, not only in configuration.

2. Topology Summary

The network consists of two redundant core routers (R1, R2), two distribution switches (SW-DR1, SW-DR2) cross-connected to one another and to every access switch, and four department access switches, each dual-homed to both distribution switches. A fifth, single-homed switch link connects an isolated Monitoring Server used exclusively by the ER/ICU department.



Device	Role	Notes
R1 / R2	Core Routers	HSRP active/standby gateway pair, OSPF adjacency over two dedicated links, DHCP server (R1)
SW-DR1 / SW-DR2	Distribution Switches	Cross-connected to all access switches and to each other; carries the intentional Layer 2 loop managed by STP
ADMIN / GUEST / ER-ICU / RADIOLOGY	Access Switches	One per department, dual-homed to both distribution switches
Monitoring Server	Application Server	HTTP dashboard and DNS service, reachable only from ER/ICU

3. IP Addressing (VLSM)

Subnets were sized to the actual host requirement of each department using VLSM, avoiding both address waste and undersized allocations.

Segment	VLAN	CIDR	Network	Usable Range
Guest WiFi	10	/25	10.10.0.0	.1 – .126
Admin / Records	20	/25	10.10.0.128	.129 – .254
ER / ICU	30	/26	10.10.1.0	.1 – .62
Radiology	40	/27	10.10.1.64	.65 – .94
R1↔R2 primary link	—	/30	10.10.2.0	.1 – .2
Monitoring	50	/29	10.10.2.8	.9 – .14
R1↔R2 backup link	—	/30	10.10.2.16	.17 – .18

Every VLAN gateway is an HSRP virtual IP rather than an individual router's real interface address, so a DHCP-assigned default gateway remains valid regardless of which core router is currently active.

VLAN	R1 (real)	R2 (real)	HSRP Virtual IP
10 – Guest	10.10.0.1	10.10.0.2	10.10.0.3
20 – Admin	10.10.0.129	10.10.0.130	10.10.0.131
30 – ER/ICU	10.10.1.1	10.10.1.2	10.10.1.3
40 – Radiology	10.10.1.65	10.10.1.66	10.10.1.67
50 – Monitoring	10.10.2.9	10.10.2.10	10.10.2.11

Net

R2#sh ip int br

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	10.10.2.2	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	unset	up	up
GigabitEthernet0/1.10	10.10.0.2	YES	manual	up	up
GigabitEthernet0/1.20	10.10.0.130	YES	manual	up	up
GigabitEthernet0/1.30	10.10.1.2	YES	manual	up	up
GigabitEthernet0/1.40	10.10.1.66	YES	manual	up	up
GigabitEthernet0/1.50	10.10.2.10	YES	manual	up	up
GigabitEthernet0/2	10.10.2.18	YES	manual	up	up
Loopback0	2.2.2.2	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

R2#

R1#sh ip int br

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	up	up
GigabitEthernet0/0.10	10.10.0.1	YES	manual	up	up
GigabitEthernet0/0.20	10.10.0.129	YES	manual	up	up
GigabitEthernet0/0.30	10.10.1.1	YES	manual	up	up
GigabitEthernet0/0.40	10.10.1.65	YES	manual	up	up
GigabitEthernet0/0.50	10.10.2.9	YES	manual	up	up
GigabitEthernet0/1	10.10.2.1	YES	manual	up	up
GigabitEthernet0/2	10.10.2.17	YES	manual	up	up
Loopback0	1.1.1.1	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

R1#

4. Redundancy Design

4.1 First-Hop Redundancy (HSRP)

R1 is configured as the active gateway (priority 110, preempt enabled) for all five VLANs; R2 serves as standby (default priority 100). Verified with a live failure test: R1's uplink was shut down, and after HSRP's hold-time window R2 correctly assumed the active role for every VLAN. When R1 was restored, preemption automatically returned active status to R1 with no manual intervention or host reconfiguration.

Before and After Router 1 Fails:

```
R2>en
R2#sh stand br
          P indicates configured to preempt.
          |
Interface  Grp  Pri  P State    Active        Standby        Virtual IP
Gig        1    100   Standby  10.10.0.1     local          10.10.0.3
Gig        1    100   Standby  10.10.0.129   local          10.10.0.131
Gig        1    100   Standby  10.10.1.1     local          10.10.1.3
Gig        1    100   Standby  10.10.1.65    local          10.10.1.67
Gig        0    100   Standby  10.10.2.9     local          10.10.2.11
R2#
%HSRP-6-STATECHANGE: GigabitEthernet0/1.30 Grp 1 state Standby -> Active
%HSRP-6-STATECHANGE: GigabitEthernet0/1.50 Grp 0 state Standby -> Active
%HSRP-6-STATECHANGE: GigabitEthernet0/1.40 Grp 1 state Standby -> Active
%HSRP-6-STATECHANGE: GigabitEthernet0/1.20 Grp 1 state Standby -> Active
%HSRP-6-STATECHANGE: GigabitEthernet0/1.10 Grp 1 state Standby -> Active
R2#sh stand br
          P indicates configured to preempt.
          |
Interface  Grp  Pri  P State    Active        Standby        Virtual IP
Gig        1    100   Active   local          unknown        10.10.0.3
Gig        1    100   Active   local          unknown        10.10.0.131
Gig        1    100   Active   local          unknown        10.10.1.3
Gig        1    100   Active   local          unknown        10.10.1.67
Gig        0    100   Active   local          unknown        10.10.2.11
R2#

R1>en
R1#sh stand br
          P indicates configured to preempt.
          |
Interface  Grp  Pri  P State    Active        Standby        Virtual IP
Gig        1    110 P Active   local          10.10.0.2     10.10.0.3
Gig        1    110 P Active   local          10.10.0.130   10.10.0.131
Gig        1    110 P Active   local          10.10.1.2     10.10.1.3
Gig        1    110 P Active   local          10.10.1.66    10.10.1.67
Gig        0    110 P Active   local          10.10.2.10    10.10.2.11
R1#
```

Router 1 restored and reclaimed Active status automatically:

```
%HSRP-6-STATECHANGE: GigabitEthernet0/1.10 Grp 1 state Speak -> Standby
%HSRP-6-STATECHANGE: GigabitEthernet0/1.30 Grp 1 state Speak -> Standby
%HSRP-6-STATECHANGE: GigabitEthernet0/1.40 Grp 1 state Speak -> Standby
%HSRP-6-STATECHANGE: GigabitEthernet0/1.20 Grp 1 state Speak -> Standby
%HSRP-6-STATECHANGE: GigabitEthernet0/1.50 Grp 0 state Speak -> Standby

R2#sh stand br
                P indicates configured to preempt.
                |
Interface    Grp  Pri  P State      Active      Standby      Virtual IP
Gig          1    100      Standby  10.10.0.1    local        10.10.0.3
Gig          1    100      Standby  10.10.0.129 local        10.10.0.131
Gig          1    100      Standby  10.10.1.1    local        10.10.1.3
Gig          1    100      Standby  10.10.1.65   local        10.10.1.67
Gig          0    100      Standby  10.10.2.9    local        10.10.2.11
R2#|

R1(config-if)#
%HSRP-6-STATECHANGE: GigabitEthernet0/0.10 Grp 1 state Standby -> Active
%HSRP-6-STATECHANGE: GigabitEthernet0/0.30 Grp 1 state Standby -> Active
%HSRP-6-STATECHANGE: GigabitEthernet0/0.50 Grp 0 state Standby -> Active
%HSRP-6-STATECHANGE: GigabitEthernet0/0.40 Grp 1 state Standby -> Active
%HSRP-6-STATECHANGE: GigabitEthernet0/0.20 Grp 1 state Standby -> Active

R1(config-if)#do sh stand br
                P indicates configured to preempt.
                |
Interface    Grp  Pri  P State      Active      Standby      Virtual IP
Gig          1    110 P Active     local        10.10.0.2    10.10.0.3
Gig          1    110 P Active     local        10.10.0.130  10.10.0.131
Gig          1    110 P Active     local        10.10.1.2    10.10.1.3
Gig          1    110 P Active     local        10.10.1.66   10.10.1.67
Gig          0    110 P Active     local        10.10.2.10   10.10.2.11
R1(config-if)#
```

4.2 Loop Prevention (Rapid-PVST)

SW-DR1 and SW-DR2 are cross-connected both directly and through every access switch, forming a deliberate Layer 2 loop. Rapid-PVST spanning tree was verified, via show spanning-tree output, to place one redundant port in Alternate/Blocking state while keeping the rest of the tree forwarding — preventing a broadcast storm while preserving the backup path. PortFast was additionally enabled on all host-facing access ports to remove unnecessary startup delay, while remaining off on every switch-to-switch and switch-to-router trunk link.

```
SW-DR1>en
SW-DR1#sh spanning-tree vlan 20
VLAN0020
  Spanning tree enabled protocol rstp
    Root ID    Priority    32788
              Address     000D.BD38.2EE2
              Cost        19
              Port        2 (FastEthernet0/2)
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID  Priority    32788  (priority 32768 sys-id-ext 20)
              Address     0060.3E8B.8015
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/2          Root FWD 19        128.2   P2p
Fa0/1          Desg FWD 19        128.1   P2p
Fa0/6          Desg FWD 19        128.6   P2p

SW-DR1#
```

```

SW-DR2>en
SW-DR2#sh spanning-tree vlan 20
VLAN0020
  Spanning tree enabled protocol rstp
    Root ID    Priority    32788
              Address     000D.BD38.2EE2
              Cost        19
              Port        5(FastEthernet0/5)
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID  Priority    32788 (priority 32768 sys-id-ext 20)
              Address     00E0.B09B.75B6
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Desg FWD 19        128.1   P2p
Fa0/6          Altn BLK 19        128.6   P2p
Fa0/5          Root FWD 19        128.5   P2p

```

SW-DR2#

```

ADMIN-SW>en
ADMIN-SW#sh spanning-tree vlan 20
VLAN0020
  Spanning tree enabled protocol rstp
    Root ID    Priority    32788
              Address     000D.BD38.2EE2
              This bridge is the root
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID  Priority    32788 (priority 32768 sys-id-ext 20)
              Address     000D.BD38.2EE2
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Desg FWD 19        128.1   P2p
Fa0/2          Desg FWD 19        128.2   P2p
Fa0/3          Desg FWD 19        128.3   P2p
Fa0/4          Desg FWD 19        128.4   P2p
Fa0/5          Desg FWD 19        128.5   P2p

```

ADMIN-SW#

4.3 Dynamic Routing and Floating Static Backup

OSPF runs only on the two dedicated R1↔R2 links, deliberately excluded from every VLAN-facing interface via passive-interface so that adjacency reflects a genuine router-to-router relationship rather than an incidental one formed over shared LAN segments. Stable router IDs are guaranteed through loopback interfaces, independent of any physical interface's state. A second physical link between R1 and R2 carries a floating static route (administrative distance 120, above OSPF's 110), confirmed by direct test to remain dormant until the primary OSPF-routed link fails, at which point it takes over automatically and traffic continues without interruption.

```
07:57:43: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/1 from FULL to DOWN, Neighbor Down: Interface down or detached
R1(config-if)#do show ip route 2.2.2.2
Routing entry for 2.2.2.2/32
Known via "static", distance 120, metric 0
  Routing Descriptor Blocks:
    * 10.10.2.18
      Route metric is 0, traffic share count is 1
R1(config-if)#
R1(config-if)#do show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
2.2.2.2          1     FULL/DR         00:00:39    10.10.2.2      GigabitEthernet0/1
R1(config-if)#
R1(config-if)#show ip route 2.2.2.2
^
% Invalid input detected at '^' marker.

R1(config-if)#do show ip route 2.2.2.2
Routing entry for 2.2.2.2/32
Known via "ospf 1", distance 110, metric 2, type intra area
  Last update from 10.10.2.2 on GigabitEthernet0/1, 00:39:14 ago
  Routing Descriptor Blocks:
    * 10.10.2.2, from 2.2.2.2, 00:39:14 ago, via GigabitEthernet0/1
      Route metric is 2, traffic share count is 1
R1(config-if)#
```

5. Security Segmentation

Three distinct access-control patterns were implemented across five Extended ACLs, each solving a different real-world segmentation problem rather than applying one uniform rule everywhere.

Rule	Type	Behavior
ER/ICU ↔ Monitoring	Full two-way lockdown	ER/ICU may only reach the monitoring server; the server may only reply to ER/ICU. Every other VLAN is explicitly denied access to the server.
Radiology → Admin/Records	One-way initiation	Radiology may initiate contact with Admin; Admin may not initiate contact with Radiology. Reply traffic is explicitly permitted.
Guest WiFi	Outbound isolation	Guest is denied from initiating contact with any internal department VLAN.
Management access (SSH)	Source-restricted	Only the Admin/Records subnet may open an SSH session to either core router, enforced by a Standard ACL applied to the VTY lines.

All Extended ACLs are replicated identically on both R1 and R2 so that department-level security policy does not depend on which router is currently the active HSRP gateway.

RD-PC1

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.0.132

Pinging 10.10.0.132 with 32 bytes of data:

Reply from 10.10.0.132: bytes=32 time<1ms TTL=127
Reply from 10.10.0.132: bytes=32 time=2ms TTL=127
Reply from 10.10.0.132: bytes=32 time<1ms TTL=127
Reply from 10.10.0.132: bytes=32 time=10ms TTL=127

Ping statistics for 10.10.0.132:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>ping 10.10.2.12

Pinging 10.10.2.12 with 32 bytes of data:

Reply from 10.10.1.65: Destination host unreachable.
Reply from 10.10.1.65: Destination host unreachable.
Reply from 10.10.1.65: Destination host unreachable.
Reply from 10.10.1.65: Destination host unreachable.

Ping statistics for 10.10.2.12:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.10.0.4

Pinging 10.10.0.4 with 32 bytes of data:

Reply from 10.10.0.4: bytes=32 time<1ms TTL=127
Reply from 10.10.0.4: bytes=32 time<1ms TTL=127
Reply from 10.10.0.4: bytes=32 time=1ms TTL=127
Reply from 10.10.0.4: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

ER-PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.2.12

Pinging 10.10.2.12 with 32 bytes of data:

Reply from 10.10.2.12: bytes=32 time<1ms TTL=127
Reply from 10.10.2.12: bytes=32 time=10ms TTL=127
Reply from 10.10.2.12: bytes=32 time<1ms TTL=127
Reply from 10.10.2.12: bytes=32 time=1ms TTL=127

Ping statistics for 10.10.2.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 10.10.0.132

Pinging 10.10.0.132 with 32 bytes of data:

Reply from 10.10.1.1: Destination host unreachable.
Reply from 10.10.1.1: Destination host unreachable.
Reply from 10.10.1.1: Destination host unreachable.
Reply from 10.10.1.1: Destination host unreachable.

Ping statistics for 10.10.0.132:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.10.0.4

Pinging 10.10.0.4 with 32 bytes of data:

Reply from 10.10.1.1: Destination host unreachable.
Reply from 10.10.1.1: Destination host unreachable.
Reply from 10.10.1.1: Destination host unreachable.
Reply from 10.10.1.1: Destination host unreachable.

Ping statistics for 10.10.0.4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Activate Windows

ADM-PC3

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.1.68

Pinging 10.10.1.68 with 32 bytes of data:

Reply from 10.10.0.129: Destination host unreachable.
Reply from 10.10.0.129: Destination host unreachable.
Reply from 10.10.0.129: Destination host unreachable.
Reply from 10.10.0.129: Destination host unreachable.

Ping statistics for 10.10.1.68:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.10.2.12

Pinging 10.10.2.12 with 32 bytes of data:

Reply from 10.10.0.129: Destination host unreachable.
Reply from 10.10.0.129: Destination host unreachable.
Reply from 10.10.0.129: Destination host unreachable.
Reply from 10.10.0.129: Destination host unreachable.

Ping statistics for 10.10.2.12:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Command Prompt

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 10.10.0.4

Pinging 10.10.0.4 with 32 bytes of data:

Reply from 10.10.2.9: Destination host unreachable.
Reply from 10.10.2.9: Destination host unreachable.
Reply from 10.10.2.9: Destination host unreachable.
Reply from 10.10.2.9: Destination host unreachable.

Ping statistics for 10.10.0.4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.10.1.68

Pinging 10.10.1.68 with 32 bytes of data:

Reply from 10.10.2.9: Destination host unreachable.
Reply from 10.10.2.9: Destination host unreachable.
Reply from 10.10.2.9: Destination host unreachable.
Reply from 10.10.2.9: Destination host unreachable.

Ping statistics for 10.10.1.68:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.10.1.6

Pinging 10.10.1.6 with 32 bytes of data:

Reply from 10.10.1.6: bytes=32 time<1ms TTL=127
Reply from 10.10.1.6: bytes=32 time=2ms TTL=127
Reply from 10.10.1.6: bytes=32 time<1ms TTL=127
Reply from 10.10.1.6: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.1.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>|
```

6. Verification and Testing

Every design claim in this document was verified against live show-command output or a simulated failure, rather than assumed from configuration alone.

- HSRP failover: R1 shut down → R2 confirmed Active for all five VLANs → R1 restored → preemption reclaimed Active status automatically.
- Security policy under failover: with R2 acting as the active gateway, all five ACL rules — including the SSH restriction — were re-tested and confirmed to hold exactly as they do under normal operation.
- STP loop prevention: show spanning-tree vlan 20 confirmed one port in Alternate/Blocking state across the redundant SW-DR1–SW-DR2 path.
- Floating static route failover: with the primary R1↔R2 link shut down, the OSPF-learned route to R2's loopback was confirmed to disappear from the routing table and be replaced by the static route (AD 120); connectivity was confirmed unbroken throughout.

7. Notable Troubleshooting

Two issues surfaced during testing that are worth documenting, since they reflect genuine debugging rather than a pre-scripted build.

7.1 ACL blocking HSRP's own control-plane traffic

The ER/ICU Extended ACL denied all traffic sourced from the ER/ICU subnet except toward the monitoring server. Because R1 and R2's own real interface addresses on that VLAN fall inside that same subnet range, their HSRP Hello multicasts were being silently filtered by the same rule intended for host traffic — causing R1 and R2 to lose sight of each other on that VLAN and both claim Active status simultaneously. This was resolved by explicitly permitting UDP port 1985 (HSRP) between the routers' real addresses ahead of the broader deny rule.

7.2 Asymmetric ACL logic blocking reply traffic

Early versions of the monitoring-server and Guest-isolation ACLs denied traffic by subnet in one direction only, which inadvertently caught legitimate return traffic — for example, a ping reply sourced from the “blocked” side of a permitted conversation. This was resolved by explicitly permitting ICMP echo-reply where two-way communication was intended, while keeping new inbound connection attempts denied, approximating one-way access control with stateless ACLs.

8. Technologies Demonstrated

VLSM subnetting · HSRP first-hop redundancy · Rapid-PVST spanning tree with PortFast · 802.1Q VLAN trunking · OSPF with passive-interface design · floating static routing · multi-pool DHCP · Standard and Extended ACLs · SSH hardening · structured failure simulation and evidence-based verification.